

ECEC state demand inputs: method

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1 Purpose

1. The simulator reads one file for each geography.
2. The file has six rows. Each row is one care type.
3. Each row holds one number. The number is the annual hours of child care.
4. This memo gives the source of that number.

Two rules apply to the whole build.

- **The build works at the state level.** It reads county data one time, in step 1, and it adds the counties together. No later step uses a county.
- **The build uses a state measurement first.** It uses a national figure only where the state publishes none.

Why the build exists

- The simulator must know the mix of child care in each state.
- NSECE measures that mix. NSECE does not record the state of a family.
- NSECE reports the Census region only.
- Therefore I build the mix from the administrative records of each state.
- I use NSECE only for the care that no administrative record shows.

2 Results

The build measures two quantities. All other rows divide these two.

	Center-based	Not center-based	Total	Share center
Kentucky	81,326	80,604	161,930	50%
New Jersey	173,225	136,721	309,946	56%
New York	270,982	378,847	649,829	42%
New York City	132,065	169,911	301,976	44%
Virginia	135,068	127,932	263,000	51%

Care that is not center-based has two parts. The build counts the first part. The build calculates the second part as a remainder.

	Licensed home care	Informal care
Kentucky	4,756	75,848
New Jersey	1,987	134,734
New York	96,366	282,481
New York City	38,642	131,269
Virginia	4,743	123,189

- New York holds 20 times more licensed home care than Kentucky or Virginia.
- This difference is correct. New York registers a home provider that keeps three children or more.
- Other states do not register a provider that small. The care exists there, but no register records it.

All figures below are counts of children. They are not record counts. They are not weighted survey figures.

	KY	NJ	NY	NYC	VA
Center-Based	37,550	116,402	168,235	86,337	87,192
Unpaid Center-Based	43,776	56,823	102,747	45,728	47,876
Paid Home-Based	4,756	1,987	96,366	38,642	4,743
Other Unpaid	45,830	81,411	170,686	79,318	74,436
Other Paid	27,363	48,607	101,909	47,357	44,442
Unpaid Home-Based	2,655	4,715	9,886	4,594	4,311
Total	161,930	309,946	649,829	301,976	263,000

The output file holds annual hours. This table shows them in millions.

	KY	NJ	NY	NYC	VA
Center-Based	63.3	196.2	283.6	145.5	147.0
Unpaid Center-Based	62.9	81.6	147.6	65.7	68.8
Paid Home-Based	9.1	3.8	185.0	74.2	9.1
Other Unpaid	88.4	157.1	329.3	153.0	143.6
Other Paid	42.5	75.5	158.4	73.6	69.1
Unpaid Home-Based	5.6	10.0	21.0	9.7	9.1

New York City has two positions in this table.

- The city is part of New York. The New York row includes it.
- The city is also a separate build.
- The two builds do not agree. Each build uses its own control total from step 2.
- Use one build or the other build. Do not add them together.

3 The build

Step 1. Count the children

1. Read the Census population estimates for 2019.
2. Keep children of age 0 to 4.
3. Add the counties together for each geography.

Children 0–4, 2019	
Kentucky	272,610
New Jersey	514,690
New York	1,127,001
New York City	523,718
Virginia	505,477

The build uses ages 0 to 4 for three reasons.

- The simulator uses the same ages. Its code keeps children below age 5 in seven places.
- A universe of ages 0 to 5 increases each total by 15 to 23 percent.
- Census publishes no county estimate for age 5 alone. The build can count a state at age 5, but it can only estimate New York City.

New York City holds 47 percent of the children in New York.

Step 2. Find the children in non-parental care

1. Read the NSCH household survey for 2019.
2. Keep the children of age 0 to 4. Weight each child by the child weight.
3. Calculate the share of children in non-parental care.
4. Multiply the children from step 1 by that share.

	Children 0–4	Rate	In non-parental care	Records
Kentucky	272,610	59.4%	161,930	147
New Jersey	514,690	60.2%	309,946	101
New York	1,127,001	57.7%	649,829	112
New York City	523,718	57.7%	301,976	112
Virginia	505,477	52.0%	263,000	121

The product is the **control total**. Every later count is a share of it.

- **The control total is always a 2019 figure.** Both inputs are 2019 data.
- This is why the build can use a register from 2026. The register sets the mix of care. The register never sets the total.

- **New York City uses the rate of New York.** NSCH publishes no rate below the state. The registers supply all city differences.
- **This rate is the weakest number in the build.** Each state has about 120 families. The standard error is about seven percentage points. Section 4 explains why the build does not pool more survey years.

Step 3. Convert the registers into children

Each state licenses child care providers. Each state publishes a register. The register gives capacity. The build needs children. Three operations convert one into the other.

Operation 1. Remove the school-age capacity. Every register includes it. Each register hides it differently.

- **Kentucky** does not mark a provider as a center or a home. The build applies the state rule: a licensed home keeps 12 children or fewer. This gives 1,609 centers and 335 homes. The build drops a provider that serves school-age children only. The build scales a mixed age band by a ratio between 0.487 and 0.827. It measures each ratio inside the register.
- **New Jersey** publishes an age range for each center. The build drops the 1,035 centers that serve ages 6 to 13 only. It scales a mixed range by 0.820.
- **New York** publishes capacity by age. Infant plus toddler plus preschool gives the under-five capacity exactly. A home reports one total, but the register publishes its school-age capacity. The build subtracts it.
- **Virginia** publishes the age range as text. The build reads the text and converts it to months. It drops a provider that starts at age 5 or later. It scales a center that serves past age 6 by 0.60.

Operation 2. Move the register back to 2019. Two registers are 2026 exports.

- **New York** scales each provider type against the state capacity series for 2015 to 2025. Group family day care scales to 0.86 statewide and to 0.84 in the city. Centers did not change. One factor for all types would make the mix incorrect.
- **Virginia** scales by one statewide factor of 0.90. The build calculates it against the state capacity report for June 2020.
- **Kentucky and New Jersey** need no correction. The Kentucky register is a snapshot from September 2018. The New Jersey register is a snapshot from May 2019.

Operation 3. Convert capacity into children. Capacity is the legal maximum. The true enrollment is lower.

- **Kentucky measured its own enrollment.** The build applies no fill rate there. The 2019 workforce study prints capacity and enrollment together: 42 children for each center and 9 for each home. This is the preferred method. It is why 1,609 centers give only 67,578 children.
- **New Jersey measured its own fill rate.** Its centers are 67 percent full. The figure comes from the market rate survey of 2021.

- **New York and Virginia publish no fill rate.** They use the national figures: 90 percent for a center and 60 percent for a home.
- New York homes use 80 percent instead. Its register already removed the school-age capacity. The 60 percent figure would remove it a second time.

		Providers	Under-5 capacity	Children
Kentucky	Centers	1,609	107,369	67,578
Kentucky	Homes	335	2,115	3,015
New Jersey	Centers	3,128	251,159	169,106
New York	Centers, state	2,073	153,053	138,191
New York	Centers, city	2,310	141,300	127,580
New York	Homes	11,918	117,401	93,381
New York City	Centers	2,310	141,300	127,580
New York City	Homes	4,580	46,057	36,634
Virginia	Centers	3,186	207,835	187,654
Virginia	Homes	1,935	12,633	7,604

New York needs two registers.

- New York has two regulators.
- The state licenses no center inside New York City.
- The city permits its own centers under Article 47 of the Health Code.
- The state register therefore shows zero centers in the city. This is correct. It is not a failure of the build.

Step 4. Add the programs that count their children

Head Start, Early Head Start and public pre-K publish their enrollment. The build uses the published count.

	Pre-K	Head Start	Early Head Start	Total
Kentucky	28,465	12,307	3,004	43,776
New Jersey	41,305	11,940	3,578	56,823
New York	54,451	36,863	11,433	102,747
New York City	26,717	13,073	5,938	45,728
Virginia	33,790	11,579	2,507	47,876

- **The build subtracts these children from the center count.** It does not add them.
- A Head Start center usually holds a state licence. Step 3 therefore counted its children already. An addition would count them two times.
- **A published count does not move in step 5.** The build treats it as a census.
- **This row is an upper limit in each state.** Head Start and state pre-K overlap. No source publishes the size of the overlap.

Step 5. Calibrate the total to the survey

A register measures the relative distribution across care types correctly. A register measures the total incorrectly. It shows too many children in formal care.

1. Take the share of non-parental care that is formal. NSECE gives this share for each Census region. It is a little more than one half.
2. Multiply that share by the control total from step 2. This gives the target.
3. Hold the program counts from step 4 constant.
4. Scale all other counts by one factor until the formal total reaches the target.

All scaled counts move together. The relative distribution does not change. Only the total changes.

	Scale factor	Formal	Informal	Informal share
Kentucky	1.58	86,082	75,848	47%
New Jersey	1.04	175,212	134,734	44%
New York	1.03	367,348	282,481	44%
New York City	1.05	170,707	131,269	44%
Virginia	0.62	139,811	123,189	47%

This table is the primary check on the build. Read it as follows.

- **New Jersey, New York and the city change very little.** Their register counts came within two percentage points of the survey before the build changed anything. Three separate state registers agree with one independent national survey. This is the strongest evidence that step 3 is correct.
- **The build reduces Virginia by one third.** Its register is the most complete of the four. It lists unlicensed and voluntarily registered family homes. Other states do not record those providers. Virginia also uses the borrowed national fill rate.
- **The build increases Kentucky by one half.** Kentucky is the only state that measured how full its centers are. The measurement is 57 percent. New York and Virginia use 90 percent. The national figure is too high by more than one half. Kentucky moves up and Virginia moves down, which is the result that this difference predicts. The calibration therefore corrects a measured error.

Step 6. Complete the rows

The build uses no price split.

- An earlier version divided center care into low-priced and high-priced.
- A state publishes one price. Each state therefore fell on one side of the national boundary.
- A care type with no children then reads as unmeasured, not as absent.
- Center care is now one row. It holds 1,686 hours for each child each year. This is the national mean across both price bands.

The build calculates the care that no register records.

- Three of the six types appear in no register: unpaid relatives, nannies and informal arrangements.
- Unpaid relative care is not a business. It files no return.
- The build subtracts all counted care from the control total. The remainder is this care.
- The build divides the remainder three ways on the national pattern.
- The remainder is 44 to 47 percent of non-parental care. The national figure is 47 percent.

The build converts children into hours.

- Each care type has a measured hours figure from NSECE.
- The figure is the mean weekly hours of the primary arrangement, multiplied by 50 weeks.
- The figures run from about 1,440 hours for program center care to about 2,120 hours for unpaid home care.
- The hours column is therefore not proportional to the children column.

4 The source of each number

	Source	Reliability
Children 0–4	Census, 2019	Counted
Share in non-parental care	Household survey, 2019, about 120 families for each state	Weak. The build tested a pool of survey years and rejected it
Provider capacity	State register	Counted, but it is a legal maximum. Virginia providers report that they can serve 27 percent fewer children than their licence permits
Fill rate, KY and NJ	The state’s own survey	Measured in the state
Fill rate, NY and VA	National survey	Borrowed. Two states report that it is too high
Program enrollment	Federal and state reports	Counted
Informal split and hours	National survey	Borrowed

Why the rate uses one survey year

A pool of survey years reduces the sampling error by 40 to 47 percent. The build does not pool. Every input must match the 2019 base year of the simulator.

1. A pool of 2018, 2019 and 2020 centres exactly on 2019. This pool passes the vintage test.
2. This pool fails on content. The survey collected the 2020 data during the pandemic. The use of child care fell.
3. A pool of 2018 and 2019 centres on the middle of 2018. Every other input is 2019.

	2018	2019	2020
Kentucky	57.3%	59.4%	45.8%
New Jersey	58.3%	60.2%	42.6%
New York	64.7%	57.7%	62.1%
Virginia	61.7%	52.0%	46.7%

The fall in 2020 is a real change. It is not sampling noise. A pool would decrease each total by several percentage points. The rate therefore keeps its full sampling error. The repository holds both extra survey years, so another person can repeat this test.

Why the borrowed fill rates are too high

Two states published their own figures. Both point in the same direction.

- **Kentucky** measured its centers at 57 percent full in 2019. New York and Virginia use 90 percent.
- **Virginia** measured a related quantity in 2022. It asked 1,968 providers what they can serve, and compared the answer to their licence. Centers reported 73 percent of their licensed capacity. Homes reported 91 percent. The sample holds 121,260 licensed places against 89,374 children.

The build uses neither figure as a rate.

- The Kentucky figure applies to Kentucky.
- The Virginia figure is from 2022. It measures capacity, not enrollment. It would duplicate the correction that the fill rate already makes.
- Both figures show the direction and the size of the error. Licensed capacity is too high by one quarter to one third. The national fill rate does not remove enough of it.

5 Limits

1. **The build cannot see unpaid relative care.** This is the largest informal category. The figure is a remainder, not a measurement. No administrative source would correct it. Only a restricted survey file would.
2. **The mix carries the date of the register.** The total is 2019 by construction. The division across care types comes from the last publication of the state. The mix changes over time. A register from 2026 shows too few family homes for 2019.
3. **Unpaid Center-Based is an upper limit in each state.** Head Start and state pre-K overlap. A program site can also hold a state licence.

4. **Step 5 forces an exact match to the survey.** This is a decision, not a result. The alternative is to report the disagreement between the register and the survey.
5. **The simulator expects seven rows.** It divides center care by price, and it rejects a file with six rows. The bridge between the two is one mechanical operation. It divides the center row on a national share. The build keeps this operation outside the method, so that the build does not appear to measure a quantity that it cannot measure.